# Advance Python Assignment

## Topic-5: Exception Handling

### Q1. Introduction to exception handling using try, except, and finally.

Ans.
Exception handling in Python is used to handle errors that occur during program execution. It helps prevent the program from crashing and allows smooth execution. The try block is used to write code that may cause an error, while the except block handles the error. The finally block is used to execute code that should run whether an error occurs or not. Exception handling makes programs more reliable and user-friendly by handling unexpected situations properly.

Example:
```
try:
   a = int(input("Enter number: "))
   b = int(input("Enter number: "))
   print(a / b)
except ZeroDivisionError:
   print("Cannot divide by zero")
finally:
   print("Execution completed")
```

### Q2. Handling multiple and custom exceptions in Python.

Ans.
In Python, we can handle multiple exceptions using multiple except blocks. This allows the program to respond differently to different types of errors. Custom exceptions are user-defined exceptions that are created to handle specific situations. They are useful when built-in exceptions are not enough for our requirements. Handling multiple and custom exceptions improves program control and makes error handling more specific and effective.

Example:
```
try:
   a = int(input("Enter number: "))
   print(10 / a)
```

```
except ZeroDivisionError:
    print("Division by zero error")
except ValueError:
    print("Invalid input")

# Custom Exception
class MyError(Exception):
    pass

try:
    raise MyError("Custom error occurred")
except MyError as e:
    print(e)
```